

Botany

Section-A

Biological Classification

(Kingdom Fungi)

- Which one of the following is not a criterion for classification of fungi?
 - Mode of spore formation
 - Fruiting body
 - Morphology of mycelium
 - Mode of nutrition
- Match List-I with List-II.

List-I		List-II	
(A)	<i>Rhizopus</i>	(I)	Mushroom
(B)	<i>Ustilago</i>	(II)	Smut fungus
(C)	<i>Puccinia</i>	(III)	Bread mould
(D)	<i>Agaricus</i>	(IV)	Rust fungus

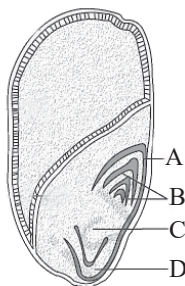
Choose the correct answer from the options given below:

- A-(III), B-(II), C-(I), D-(IV)
- A-(IV), B-(III), C-(II), D-(I)
- A-(III), B-(II), C-(IV), D-(I)
- A-(I), B-(III), C-(II), D-(IV)

Morphology of Flowering Plants

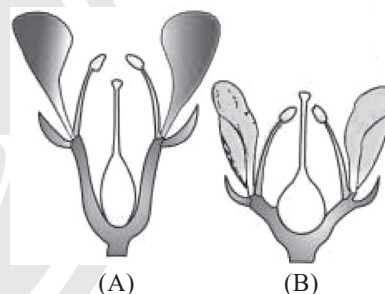
(The Seed)

- Identify the part of the seed from the given figure which is destined to form root when the seed germinates.
 - C
 - D
 - A
 - B



(The Flower)

- Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (A) and (B).



- (A) Perigynous; (B) Epigynous
 - (A) Perigynous; (B) Perigynous
 - (A) Epigynous; (B) Hypogynous
 - (A) Hypogynous; (B) Epigynous
- Which of the following is an example of actinomorphic flower?
 - Pisum*
 - Sesbania*
 - Datura*
 - Cassia*

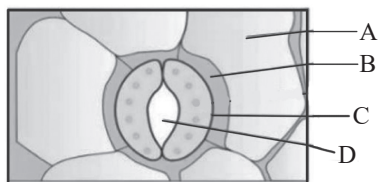
Anatomy of Flowering Plants

(Anatomy of Dicot & Monocot Plants)

- Bulliform cells are responsible for
 - Increased photosynthesis in monocots.
 - Providing large spaces for storage of sugars.
 - Inward curling of leaves in monocots.
 - Protecting the plant from salt stress.

(Tissue System)

7. In the given figure, which component has thin outer walls and highly thickened inner walls?



- a. A b. B
c. C d. D

(Permanent Tissues)

8. Given below are two statements:

Statement I: Parenchyma is living but collenchyma is dead tissue.

Statement II: Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperms.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is true but Statement II is false
b. Statement I is false but Statement II is true
c. Both Statement I and Statement II are true
d. Both Statement I and Statement II are false

Cell The Unit of Life

(Nucleus and Microbodies)

9. Match List-I with List-II.

List-I		List-II	
(A)	Nucleolus	(I)	Site of formation of glycolipid
(B)	Centriole	(II)	Organization like the cartwheel
(C)	Leucoplasts	(III)	Site for active ribosomal RNA synthesis
(D)	Golgi apparatus	(IV)	For storing nutrients

Choose the correct answer from the options given below:

- a. A-(III), B-(IV), C-(II), D-(I)
b. A-(I), B-(II), C-(III), D-(IV)
c. A-(III), B-(II), C-(IV), D-(I)
d. A-(II), B-(III), C-(I), D-(IV)

Biomolecules

(Enzymes)

10. Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of:
- a. Competitive inhibition b. Enzyme activation
c. Cofactor inhibition d. Feedback inhibition
11. The cofactor of the enzyme carboxypeptidase is
- a. Flavin b. Haem
c. Zinc d. Niacin

(Analysis of Chemical Composition)

12. Lecithin, a small molecular weight organic compound found in living tissues, is an example of:
- a. Glycerides b. Carbohydrates
c. Amino acids d. Phospholipids

Cell Cycle and Cell Division

(Mitosis)

13. Spindle fibers attach to kinetochores of chromosomes during
- a. Anaphase b. Telophase
c. Prophase d. Metaphase

(Meiosis)

14. Given below are two statements:

Statement I: Chromosomes become gradually visible under light microscope during leptotene stage.

Statement II: The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is true but Statement II is false
b. Statement I is false but Statement II is true
c. Both Statement I and Statement II are true
d. Both Statement I and Statement II are false

Photosynthesis in Higher Plants

(Where are ATP & NADPH Used?)

15. Which of the following are required for the dark reaction of photosynthesis?
- A. Light B. Chlorophyll
C. CO₂ D. ATP
E. NADPH

Choose the correct answer from the options given below:

- a. C, D and E only b. D and E only
c. A, B and C only d. B, C and D only
16. How many molecules of ATP and NADPH are required for every molecule of CO₂ fixed in the Calvin cycle?
- a. 3 molecules of ATP and 3 molecules of NADPH
b. 3 molecules of ATP and 2 molecules of NADPH
c. 2 molecules of ATP and 3 molecules of NADPH
d. 2 molecules of ATP and 2 molecules of NADPH

Plant Growth and Development

(Differentiation, Dedifferentiation, Redifferentiation and Development)

17. Formation of interfascicular cambium from fully developed parenchyma cells is an example for
- a. Dedifferentiation b. Maturation
c. Differentiation d. Redifferentiation

(Plant Growth Regulators)

18. Auxin is used by gardeners to prepare weed free lawns. But no damage is caused to grass as auxin
- does not affect mature monocotyledonous plants.
 - can help in cell division in grasses, to produce growth.
 - promotes apical dominance.
 - promotes abscission of mature leaves only.

Sexual Reproduction in Flowering Plants

(Flowering and Pre-fertilisation)

19. Identify the set of correct statements:
- The flowers of *Vallisneria* are colourful and produce nectar.
 - The flowers of waterlily are not pollinated by water.
 - In most of water-pollinated species, the pollen grains are protected from wetting.
 - Pollen grains of some hydrophytes are long and ribbon like.
 - In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below:

- A, C, D and E only
- B, C, D and E only
- C, D and E only
- A, B, C and D only

Principles of Inheritance and Variation

(Mendel's Laws of Inheritance And Inheritance of One Gene)

20. In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?
- Bb
 - BB/Bb
 - BB
 - bb
21. A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?
- Only pink flowered plants
 - Red, Pink as well as white flowered plants
 - Only red flowered plants
 - Red flowered as well as pink flowered plants

22. Match List-I with List-II.

List-I		List-II	
(A)	Two or more alternative forms of a gene	(I)	Back cross
(B)	Cross of F_1 progeny with homozygous recessive parent	(II)	Ploidy
(C)	Cross of F_1 progeny with any of the parents	(III)	Allele
(D)	Number of chromosome sets in plant	(IV)	Test cross

Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(I), D-(II)
 - A-(IV), B-(III), C-(III), D-(I)
 - A-(I), B-(II), C-III, D-(IV)
 - A-(II), B-(I), C-(III), D-(IV)
23. Which one of the following can be explained on the basis of Mendel's Law of Dominance?
- Out of one pair of factors one is dominant and the other is recessive.
 - Alleles do not show any expression and both the characters appears as such in F_2 generation.
 - Factors occur in pairs in normal diploid plants.
 - The discrete unit controlling a particular character is called factor.
 - The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

- B, C and D only
- A, B, C, D and E
- A, B and C only
- A, C, D and E only

Molecular Basis of Inheritance

(Transcription)

24. A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;
- Inducer, Repressor, Structural gene
 - Promotor, Structural gene, Terminator
 - Repressor, Operator gene, Structural gene
 - Structural gene, Transposons, Operator gene

(Regulation of Gene Expression)

25. The lactose present in the growth medium of bacteria is transported to the cell by the action of
- Permease
 - Polymerase
 - Beta-galactosidase
 - Acetylase

Microbes in Human Welfare

(Microbes in Household & Industrial Products)

26. Match List-I with List-II.

List-I		List-II	
(A)	<i>Clostridium butylicum</i>	(I)	Ethanol
(B)	<i>Saccharomyces cerevisiae</i>	(II)	Streptokinase
(C)	<i>Trichoderma polysporum</i>	(III)	Butyric acid
(D)	<i>Streptococcus</i> sp.	(IV)	Cyclosporin-A

Choose the correct answer from the options given below:

- A-(III), B-(I), C-(IV), D-(II)
- A-(IV), B-(I), C-(III), D-(II)
- A-(III), B-(I), C-(II), D-(IV)
- A-(II), B-(IV), C-(III), D-(I)

Biotechnology : Principles and Processes

(Tools of Recombinant DNA Technology)

27. *Hind* II always cuts DNA molecules at a particular point called recognition sequence and it consists of:

a. 4 bp
b. 10 bp
c. 8 bp
d. 6 bp

(Principles of Biotechnology)

28. What is the fate of a piece of DNA carrying a gene of interest which is transferred into an alien organism?

A. The piece of DNA would be able to multiply, itself independently in the progeny cells of the organism.
B. It may get integrated into the genome of recipient.
C. It may multiply and be inherited along with the host DNA.
D. The alien piece of DNA is not an integral part of chromosome.
E. It shows ability to replicate.

Choose the correct answer from the options given below:

a. B and C only
b. A and E only
c. A and B only
d. D and E only

Biotechnology and Its Applications

(Biotechnological Applications in Agriculture)

29. The capacity to generate a whole plant from any cell of the plant is called:

a. Differentiation
b. Somatic hybridization
c. Totipotency
d. Micropropagation

30. Given below are two statements:

Statement I: Bt toxins are insect group specific and coded by a gene *cry IAc*.

Statement II: Bt toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below:

a. Statement I is true but Statement II is false
b. Statement I is false but Statement II is true
c. Both Statement I and statement II are true
d. Both Statement I and Statement II are false

Organisms and Populations

(Population Attributes, Population Growth and Life History Variation)

31. The equation of Verhulst-Pearl logistic growth is

$$\frac{dN}{dt} = rN \left[\frac{K - N}{K} \right]$$

From this equation, *K* indicates:

a. Carrying capacity
b. Population density
c. Intrinsic rate of natural increase
d. Biotic potential

Biodiversity and Conservation

(Biodiversity Conservation)

32. The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called;

a. Semi-conservative method
b. Sustainable development
c. *in-situ* conservation
d. Biodiversity conservation

(Biodiversity & Patterns of Biodiversity)

33. Tropical regions show greatest level of species richness because

A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
B. Tropical environments are more seasonal.
C. More solar energy is available in tropics.
D. Constant environments promote niche specialization.
E. Tropical environments are constant and predictable.

Choose the correct answer from the options given below:

a. A, B and E only b. A, B and D only
c. A, C, D and E only d. A and B only

(Importance & Loss of Biodiversity)

34. These are regarded as major causes of biodiversity loss:

A. Over exploitation
B. Co-extinction
C. Mutation
D. Habitat loss and fragmentation
E. Migration

Choose the correct option:

a. A, B and E only b. A, B and D only
c. A, C and D only d. A, B, C and D only

35. List of endangered species was released by-

a. FOAM b. IUCN
c. GE AC d. WWF

Botany

Section-B

Plant Kingdom

(Algae)

36. Read the following statements and choose the set of correct statements.

In the members of Phaeophyceae,

- A. Asexual reproduction occurs usually by biflagellate zoospores.
- B. Sexual reproduction is by oogamous method only.
- C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.
- D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.
- E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

- a. A, C, D and E only
- b. A, B, C and E only
- c. A, B, C and C only
- d. B, C, D and E only

Morphology of Flowering Plants

(The Flower)

37. Match List-I with List-II.

List-I		List-II	
(A)	Rose	(I)	Twisted aestivation
(B)	Pea	(II)	Perigynous flower
(C)	Cotton	(III)	Drupe
(D)	Mango	(IV)	Marginal placentation

Choose the correct answer from the options given below:

- a. A-(IV), B-(III), C-(II), D-(I)
- b. A-(II), B-(III), C-(IV), D-(I)
- c. A-(II), B-(IV), C-(I), D-(III)
- d. A-(I), B-(II), C-(III), D-(IV)

38. Match List-I with List-II.

List-I (Types of stamen)		List-II (Example)	
(A)	Monoadelphous	(I)	Citrus
(B)	Diadelphous	(II)	Pea
(C)	Polyadelphous	(III)	Lily
(D)	Epiphyllous	(IV)	China-rose

Choose the correct answer from the option below:

- a. A-(I), B-(II), C-(IV), D-(III)
- b. A-(III), B-(I), C-(IV), D-(II)
- c. A-(IV), B-(II), C-(I), D-(III)
- d. A-(IV), B-(I), C-(II), D-(III)

Cell The Unit of Life

(Mitochondria, Plastids, Ribosomes)

39. The DNA present in chloroplast is:

- a. Linear, single stranded
- b. Circular, single stranded
- c. Linear, double stranded
- d. Circular, double stranded

Biomolecules

(Biomacromolecules, Proteins)

40. Match List-I with List-II.

List-I		List-II	
(A)	GLUT-4	(I)	Hormone
(B)	Insulin	(II)	Enzyme
(C)	Trypsin	(III)	Intercellular ground substance
(D)	Collagen	(IV)	Enables glucose transport into cells

Choose the correct answer from the options given below:

- a. A-(II) B-(III), C-(IV), D-(I)
- b. A-(III), B-(IV), C-(I), D-(II)
- c. A-(IV), B-(I), C-(II), D-(III)
- d. A-(I), B-(II), C-(III), D-(IV)

Plant Growth and Development

(Plant Growth Regulators)

41. Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield?

- a. Cytokinin
- b. Absciscic acid
- c. Auxin
- d. Gibberellin

Photosynthesis in Higher Plants

(C₄ Pathway & Photorespiration)

42. Given below are two statements:

Statement I: In C₃ plants, some O₂ binds RuBisCO, hence CO₂ fixation is decreased.

Statement II: In C₄ plants, mesophyll cells shows very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is true but Statement II is false
- b. Statement I is false but Statement II is true
- c. Both Statement I and Statement II are true
- d. Both Statement I and Statement II are false

Respiration in Plants

(Aerobic Respiration)

43. Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.
- Succinyl-CoA $\rightarrow$ Succinic acid
 - Isocitrate $\rightarrow$ α -ketoglutaric acid
 - Malic acid $\rightarrow$ Oxaloacetic acid
 - Succinic acid $\rightarrow$ Malic acid
44. Match List-I with List-II

List-I		List-II	
(A)	Citric acid cycle	(I)	Cytoplasm
(B)	Glycolysis	(II)	Mitochondrial matrix
(C)	Electron transport	(III)	Intermembrane space of mitochondria
(D)	Proton gradient	(IV)	Inner mitochondrial membrane

Choose the correct answer from the option given below:

- A-(III), B-(IV), C-(I), D-(II)
- A-(IV), B-(III), C-(II), D-(I)
- A-(I), B-(II), C-(III), D-(IV)
- A-(II), B-(I), C-(IV), D-(III)

Sexual Reproduction in Flowering Plants

(Flower and Pre-fertilisation)

45. Identify the correct description about the given figure:



- Cleistogamous flowers showing autogamy.
- Compact inflorescence showing complete autogamy.
- Wind pollinated plant inflorescence showing flowers with well exposed stamens.
- Water pollinated flowers showing stamens with mucilaginous covering.

Molecular Basis of Inheritance

(Regulation of Gene Expression)

46. Match List-I with List-II

List-I		List-II	
(A)	Frederick Griffith	(I)	Genetic code
(B)	Francois Jacob & Jacques	(II)	Semi-conservative mode of DNA replication
(C)	Har Gobind Khorana	(III)	Transformation
(D)	Meselson & Stahl	(IV)	Lac operon

Choose the correct answer from the options given below:

- A-(II), B-(III), C-(IV), D-(I)
- A-(IV), B-(I), C-(II), D-(III)
- A-(III), B-(II), C-(I), D-(IV)
- A-(III), B-(IV), C-(I), D-(II)

(Replication)

47. Which of the following statement is correct regarding the process of replication in *E.coli*?
- The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ as well as $3' \rightarrow 5'$ direction.
 - The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ direction.
 - The DNA dependent DNA polymerase catalyses polymerization in one direction that is $3' \rightarrow 5'$.
 - The DNA dependent RNA polymerase catalyses polymerization in one direction, that is $5' \rightarrow 3'$.

Biotechnology and Its Applications

(Biotechnological Applications in Agriculture)

48. Which of the following are fused in somatic hybridization involving two varieties of plants?
- Protoplasts
 - Pollens
 - Callus
 - Somatic embryos

Ecosystem

(Productivity)

49. In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is $100x \text{ (kcalm}^{-2} \text{ yr}^{-1})$, what would be the GPP (Gross Primary Productivity) of the third trophic level of the same ecosystem?
- $10x \text{ (kcalm}^{-2} \text{ yr}^{-1})$
 - $\frac{100x}{3x} \text{ (kcalm}^{-2} \text{ yr}^{-1})$
 - $\frac{x}{10} \text{ (kcalm}^{-2} \text{ yr}^{-1})$
 - $x \text{ (kcalm}^{-2} \text{ yr}^{-1})$

Biodiversity and Conservation

(Importance & Loss of Biodiversity)

50. Match List-I with List-II.

List-I		List-II	
(A)	Robert May	(I)	Species-Area relationship
(B)	Alexander von Humboldt	(II)	Long term ecosystem experiment using out door plots
(C)	Paul Ehrlich	(III)	Global species diversity at about 7 million
(D)	David Tilman	(IV)	Rivet popper hypothesis

Choose the correct answer from the options given below:

- A-(I), B-(III), C-(II), D-(IV)
- A-(III), B-(IV), C-(II), D-(I)
- A-(II), B-(III), C-(I), D-(IV)
- A-(III), B-(I), C-(IV), D-(II)

Zoology

Section-A

Animal Kingdom

(Basis of Classification)

51. Consider the following statements:

- A. Annelids are true coelomates
- B. Poriferans are pseudocoelomates
- C. Aschelminthes are acoelomates
- D. Platyhelminthes are pseudocoelomates

Choose the correct answer from the options given below :

- a. C only
- b. D only
- c. B only
- d. A only

(Class Cyclostomata, Chondrichythes, Osteichythes)

52. Match List-I with List-II:

List-I		List-II	
(A)	<i>Pleurobrachia</i>	(I)	Mollusca
(B)	Radula	(II)	Ctenophora
(C)	Stomochord	(III)	Osteichthytes
(D)	Air bladder	(IV)	Hemichordata

Choose the correct answer from the options given below:

- a. A-(II), B-(IV), C-(I), D-(III)
- b. A-(IV), B-(III), C-(II), D-(I)
- c. A-(IV), B-(II), C-(III), D-(I)
- d. A-(II), B-(I), C-(IV), D-(III)

53. Match List-I with List-II:

List-I		List-II	
(A)	<i>Pterophyllum</i>	(I)	Hag fish
(B)	<i>Myxine</i>	(II)	Saw fish
(C)	<i>Pristis</i>	(III)	Angel fish
(D)	<i>Exocoetus</i>	(IV)	Flying fish

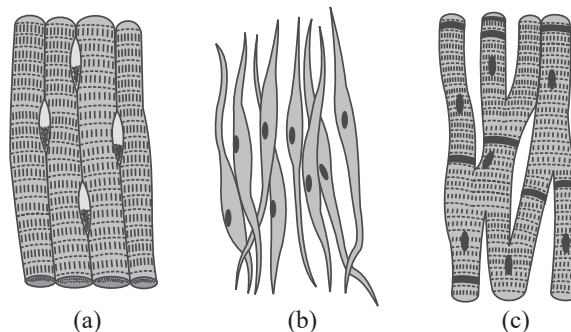
Choose the correct answer from the options given below:

- a. A-(IV), B-(I), C-(II), D-(III)
- b. A-(III), B-(II), C-(I), D-(IV)
- c. A-(II), B-(I), C-(III), D-(IV)
- d. A-(III), B-(I), C-(II), D-(IV)

Structural Organisation in Animals

(Muscle Tissue & Neural Tissue)

54. Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body:



Name of muscle/location

- a. (a) Skeletal - Biceps
- (b) Involuntary - Intestine
- (c) Smooth - Heart.
- b. (a) Involuntary - Nose tip
- (b) Skeletal - Bone
- (c) Cardiac - Heart.
- c. (a) Smooth - Toes
- (b) Skeletal - Legs
- (c) Cardiac - Heart.
- d. (a) Skeletal-Triceps
- (b) Smooth - Stomach
- (c) Cardiac - Heart.

(Cockroach)

55. In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on :

- a. 8th and 9th segment
- b. 11th segment
- c. 5th segment
- d. 10th segment

Cell: The Unit of Life

(Nucleus and Microbodies)

56. Match List-I with List-II:

List-I		List-II	
(A)	Axoneme	(I)	Centriole
(B)	Cartwheel pattern	(II)	Cilia and flagella
(C)	Crista	(III)	Chromosome
(D)	Satellite	(IV)	Mitochondria

Choose the correct answer from the options given below :

- a. A-(II), B-(IV), C-(I), D-(III)
- b. A-(II), B-(I), C-(IV), D-(III)
- c. A-(IV), B-(III), C-(II), D-(I)
- d. A-(IV), B-(II), C-(III), D-(I)

Biomolecules

(Polysaccharides, Nucleic Acids & Types of Bond)

57. Match List-I with List-II.

List-I		List-II	
(A)	Lipase	(I)	Peptide bond
(B)	Nuclease	(II)	Ester bond
(C)	Protease	(III)	Glycosidic bond
(D)	Amylase	(IV)	Phosphodiester bond

Choose the correct answer from the options given below:

- A-(II), B-(IV), C-(I), D-(III)
- A-(IV), B-(I), C-(III), D-(II)
- A-(IV), B-(II), C-(III), D-(I)
- A-(III), B-(II), C-(I), D-(IV)

Cell Cycle and Cell Division

(Cell Cycle)

58. Following are the stages of cell division :

- Gap 2 phase
- Cytokinesis
- Synthesis phase
- Karyokinesis
- Gap 1 phase

Choose the correct sequence of stages from the options given below :

- B-D-E-A-C
- E-C-A-D-B
- C-E-D-A-B
- E-B-D-A-C

(Meiosis)

59. Match List-I with List-II:

List-I (Sub Phases of Prophase I)		List-II (Specific characters)	
(A)	Diakinesis	(I)	Synaptonemal complex formation
(B)	Pachytene	(II)	Completion of terminalisation of chiasmata
(C)	Zygotene	(III)	Chromosomes look like thin threads
(D)	Leptotene	(IV)	Appearance of recombination nodules

Choose the correct answer from the options given below;

- A-(II), B-(IV), C-(I), D-(III)
- A-(IV), B-(III), C-(II), D-(I)
- A-(IV), B-(II), C-(III), D-(I)
- A-(I), B-(II), C-(IV), D-(III)

Breathing and Exchange of Gases

(Respiratory Volumes & Capacities)

60. Match List-I with List-II:

List-I		List-II	
(A)	Expiratory capacity	(I)	Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
(B)	Functional residual capacity	(II)	Tidal volume + Expiratory reserve volume
(C)	Vital capacity	(III)	Tidal volume + Inspiratory reserve volume
(D)	Inspiratory capacity	(IV)	Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :

- A-(II), B-(I), C-(IV), D-(III)
- A-(I), B-(III), C-(II), D-(IV)
- A-(II), B-(IV), C-(I), D-(III)
- A-(III), B-(II), C-(IV), D-(I)

(Exchange and Transport of Gases)

61. Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

- Low $p\text{CO}_2$ and High H^+ concentration
- Low $p\text{CO}_2$ and High temperature
- High $p\text{O}_2$ and High $p\text{CO}_2$
- High $p\text{O}_2$ and Lesser H^+ concentration

Body Fluids and Circulation

(Circulatory Pathways)

62. Following are the stages of pathway for conduction of an action potential through the heart:

- AV bundle
- Purkinje fibres
- AV node
- Bundle branches
- SA node

Choose the correct sequence of pathway from the options given below:

- B-D-E-C-A
- E-A-D-B-C
- E-C-A-D-B
- A-E-C-B-D

Excretory Products and Their Elimination

(Urine Formation, Function of Tubules)

63. Given below are two statements:

Statement I: In the nephron, the descending limb of loop of Henle impermeable to water and permeable to electrolytes.

Statement II: The proximal convoluted tubule is lined by simple columnar brush border epithelium and increases the surface area for reabsorption.

In the light of the above statements, choose the correct answer from the options given below :

- Statement I is true but Statement II is false
- Statement I is false but Statement II is true
- Both Statement I and Statement II are true
- Both Statement I and Statement II are false

Locomotion and Movement

(Joints)

64. Match List-I with List-II:

List-I		List-II	
(A)	Fibrous joints	(I)	Adjacent vertebrae, limited movement
(B)	Cartilaginous joints	(II)	Humerus and Pectoral girdle, rotational movement
(C)	Hinge joints	(III)	Skull, don't allow any movement
(D)	Ball and socket joints	(IV)	Knee, help in locomotion

Choose the correct answer from the options given below:

- A-(II), B-(III), C-(I), D-(IV)
- A-(III), B-(I), C-(IV), D-(II)
- A-(IV), B-(II), C-(III), D-(I)
- A-(I), B-(III), C-(II), D-(IV)

(Disorders of Muscular and Skeletal System)

65. Which of the following are Autoimmune disorders?

- Myasthenia gravis
- Rheumatoid arthritis
- Gout
- Muscular dystrophy
- Systemic Lupus Erythematosus (SLE)

Choose the most appropriate answer from the options given below:

- B, C & E only
- C, D & E only
- A, B & D only
- A, B & E only

Neural Control and Coordination

(Central Nervous System)

66. Match List-I with List-II:

List-I		List-II	
(A)	Pons	(I)	Provides additional space for Neurons, regulates posture and balance.
(B)	Hypothalamus	(II)	Controls respiration and gastric secretions.
(C)	Medulla	(III)	Connects different regions of the brain.
(D)	Cerebellum	(IV)	Neuro secretory cells

Choose the correct answer from the options given below:

- A-(I), B-(III), C-(II), D-(IV)
- A-(II), B-(I), C-(III), D-(IV)
- A-(II), B-(III), C-(I), D-(IV)
- A-(III), B-(IV), C-(II), D-(I)

Chemical Coordination and Integration

(Mechanism of Hormone Action)

67. Which of the following is not a steroid hormone?

- Progesterone
- Glucagon
- Cortisol
- Testosterone

Human Reproduction

(Female Reproductive System)

68. Given below are two statements:

Statement I : The presence or absence of hymen is not a reliable indicator of virginity.

Statement II : The hymen is torn during the first coitus only.

In the light of the above statements, choose the correct answer from the options given below :

- Statement I is true but Statement II is false
- Statement I is false but Statement II is true
- Both Statement I and Statement II are true
- Both Statement I and Statement II are false

69. Which of the following is not a component of Fallopian tube?

- Infundibulum
- Ampulla
- Uterine fundus
- Isthmus

(Menstrual Cycle)

70. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: FSH acts upon ovarian follicles in female and Leydig cells in male.

Reason R: Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in male human being.

In the light of the above statements, choose the correct answer from the options given below:

- A is true but R is false
- A is false but R is true
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is NOT the correct explanation of A.

(Parturition and Lactation)

71. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A: Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby.

Reason R : Colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A is correct but R is not correct.
- A is not correct but R is correct.
- Both A and R are correct and R is the correct explanation of A
- Both A and R are correct but R is NOT the correct explanation of A.

Reproductive Health

(Population Stabilisation and Birth Control)

72. Match List-I with List-II:

List-I		List-II	
(A)	Non-medicated IUD	(I)	Multiload 375
(B)	Copper releasing IUD	(II)	Progestogens
(C)	Hormone releasing IUD	(III)	Lippes loop
(D)	Implants	(IV)	LNG - 20

Choose the correct answer from the options given below:

- A-(IV), B-(I), C-(II), D-(III)
- A-(III), B-(I), C-(IV), D-(II)
- A-(III), B-(I), C-(II), D-(IV)
- A-(I), B-(III), C-(IV), D-(II)

73. Which of the following is not a natural/traditional contraceptive method?

- Lactational amenorrhea
- Vaults
- Coitus interruptus
- Periodic abstinence

Principles of Inheritance and Variation

(Genetic Disorders)

74. Match List-I with List-II:

List-I		List-II	
(A)	Down's syndrome	(I)	11 th chromosome
(B)	α -Thalassemia	(II)	'X' chromosome
(C)	β -Thalassemia	(III)	21 st chromosome
(D)	Klinefelter's syndrome	(IV)	16 th chromosome

Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(I), D-(II)
- A-(IV), B-(I), C-(II), D-(III)
- A-(I), B-(II), C-(III), D-(IV)
- A-(II), B-(III), C-(IV), D-(I)

Molecular Basis of Inheritance

(Transcription)

75. Which one is the correct product of DNA dependent RNA polymerase to the given template?

3'TACATGGCAAATATCCATTCA5'

- 5'AUGUACCGUUUAUAGGGAAGU3'
- 5'ATGTACCGTTTATAGGTAAGT3'
- 5'AUGUACCGUUUAUAGGUAAGU3'
- 5'AUGUAAAGUUUAUAGGUAAGU3'

Evolution

(Evidences of Evolution)

76. The flippers of the Penguins and Dolphins are the example of the

- Convergent evolution
- Divergent evolution
- Adaptive radiation
- Natural selection

(Hardy-Weinberg Principle)

77. Which one of the following factors will not affect the Hardy-Weinberg equilibrium?

- Gene migration
- Constant gene pool
- Genetic recombination
- Genetic drift

(Human Evolution)

78. Given below are some stages of human evolution. Arrange them in correct sequence. (Past to Recent)

- Homo habilis*
- Homo sapiens*
- Homo neanderthalensis*
- Homo erectus*

Choose the correct sequence of human evolution from the options given below:

- C-B-D-A
- A-D-C-B
- D-A-C-B
- B-A-D-C

Human Health and Diseases

(Common Diseases in Humans)

79. Match List-I with List-II:

List-I		List-II	
(A)	Typhoid	(I)	Fungus
(B)	Leishmaniasis	(II)	Nematode
(C)	Ringworm	(III)	Protozoa
(D)	Filariasis	(IV)	Bacteria

Choose the correct answer from the options given below:

- A-(III), B-(I), C-(IV), D-(II)
- A-(II), B-(IV), C-(III), D-(I)
- A-(I), B-(III), C-(II), D-(IV)
- A-(IV), B-(III), C-(I), D-(II)

80. Match List-I with List-II:

List-I		List-II	
(A)	Common cold	(I)	<i>Plasmodium</i>
(B)	Haemozoin	(II)	Typhoid
(C)	Widal test	(III)	<i>Rhinoviruses</i>
(D)	Allergy	(IV)	Dust mites

Choose the correct answer from the options given below:

- A-(III), B-(I), C-(II), D-(IV)
- A-(IV), B-(II), C-(III), D-(I)
- A-(II), B-(IV), C-(III), D-(I)
- A-(I), B-(III), C-(II), D-(IV)

(Drugs & Alcohol Abuse)

81. Match List-I with List-II:

List-I		List-II	
(A)	Cocaine	(I)	Effective sedative in surgery
(B)	Heroin	(II)	<i>Cannabis sativa</i>
(C)	Morphine	(III)	<i>Erythroxylum</i>
(D)	Marijuana	(IV)	<i>Papaver somniferum</i>

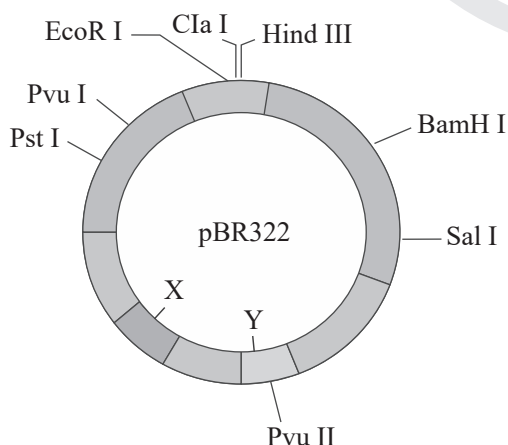
Choose the correct answer from the options given below :

- A-(II), B-(I), C-(III), D-(IV)
- A-(III), B-(IV), C-(I), D-(II)
- A-(IV), B-(III), C-(I), D-(II)
- A-(I), B-(III), C-(II), D-(IV)

Biotechnology: Principles and Processes

(Tools of Recombinant DNA Technology)

82. The following diagram showing restriction sites in *E. coli* cloning vector pBR322. Find the role of 'X' and 'Y' genes :



- The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.
- Gene 'X' is responsible for recognition sites and 'Y' is responsible for antibiotic resistance.
- The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.
- The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.

83. The "Ti plasmid" of *Agrobacterium tumefaciens* stands for

- Tumor inducing plasmid
- Temperature independent plasmid
- Tumour, inhibiting plasmid
- Tumor independent plasmid

(Processes of Recombinant DNA Technology)

84. Which of the following statements is incorrect?

- Bio-reactors are used to produce small scale bacterial cultures.
- Bio-reactors have an agitator system, an oxygen delivery system and foam control system.
- A bio-reactor provides optimal growth conditions for achieving the desired product.
- Most commonly used bio-reactors are of stirring type.

Biotechnology and Its Applications

(Transgenic Animals)

85. Match List-I with List-II:

List-I		List-II	
(A)	α - 1 antitrypsin	(I)	Cotton bollworm
(B)	<i>Cry IAb</i>	(II)	ADA deficiency
(C)	<i>Cry IAc</i>	(III)	Emphysema
(D)	Enzyme replacement therapy	(IV)	Corn borer

Choose the correct answer from the options given below :

- A-(III), B-(IV), C-(I), D-(II)
- A-(II), B-(IV), C-I D-(III)
- A-(II), B-(I), C-(IV), D-(III)
- A-(III), B-(I), C-(II), D-(IV)

Zoology

Section-B

Animal Kingdom

(Hemichordata and Chordata)

86. The following are the statements about non-chordates:

- A. Pharynx is perforated by gill slits.
- B. Notochord is absent.
- C. Central nervous system is dorsal.
- D. Heart is dorsal if present.
- E. Post anal tail is absent.

Choose the most appropriate answer from the options given below:

- a. B, D & E only
- b. B, C & D only
- c. A & C only
- d. A, B & D only

Structural Organisation in Animals

(Epithelial & Connective Tissue)

87. Match List-I with List-II:

List-I		List-II	
(A)	Unicellular glandular epithelium	(I)	Salivary glands
(B)	Compound epithelium	(II)	Pancreas
(C)	Multicellular glandular epithelium	(III)	Goblet cells of alimentary canal
(D)	Endocrine glandular epithelium	(IV)	Moist surface of buccal cavity

Choose the correct answer from the options given below:

- a. A-(III), B-(IV), C-(I), D-(II)
- b. A-(II), B-(I), C-(IV), D-(III)
- c. A-(II), B-(I), C-(III), D-(IV)
- d. A-(IV), B-(III), C-(I), D-(II)

(Cockroach)

88. Match List-I with List-II related to digestive system of cockroach.

List-I		List-II	
(A)	The structures used for storing of food.	(I)	Gizzard
(B)	Ring of 6-8 blind tubules at junction of foregut and midgut.	(II)	Gastric Caeca
(C)	Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	(III)	Malpighian tubules
(D)	The structures used for grinding the food	(IV)	Crop

Choose the correct answer from the options given below:

- a. A-(IV), B-(III), C-(II), D-I
- b. A-(III), B-(II), C-(IV), D-I
- c. A-(IV), B-(II), C-(III), D-I
- d. A-(I), B-(II), C-(III), D-IV

Cell: The Unit of Life

(Mitochondria, Plastids, Ribosomes)

89. Given below are two statements:

Statement I: Mitochondria and chloroplasts are both double membrane bound organelles.

Statement II: Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.

In the light of the above statements, choose the most appropriate answer from the options given below:

- a. Statement I is correct but Statement II is incorrect.
- b. Statement I is incorrect but Statement II is correct.
- c. Both Statement I and Statement II are correct.
- d. Both Statement I and Statement II are incorrect.

Biomolecules

(Enzymes)

90. Regarding catalytic cycle of an enzyme action select the correct sequential steps :

- A. Substrate enzyme complex formation.
- B. Free enzyme ready to bind with another substrate.
- C. Release of products.
- D. Chemical bonds of the substrate broken.
- E. Substrate binding to active site.

Choose the correct answer from the options given below :

- a. B, A, C, D, E
- b. E, D, C, B, A
- c. E, A, D, C, B
- d. A, E, B, D, C

Body Fluids and Circulation

(Circulatory Pathways)

91. Match List-I with List-II:

List-I		List-II	
(A)	P wave	(I)	Heart muscles are electrically silent.
(B)	QRS complex	(II)	Depolarisation of ventricles.
(C)	T wave	(III)	Depolarisation of atria.
(D)	T-P gap	(IV)	Repolarisation of ventricles.

Choose the correct answer from the options given below:

- A-(II), B-(III), C-(I), D-(IV)
- A-(IV), B-(II), C-(I), D-(III)
- A-(I), B-(III), C-(IV), D-(II)
- A-(III), B-(II), C-(IV), D-(I)

Excretory Products and Their Elimination

(Excretory Organs, Human Excretory System)

92. Choose the correct statement given below regarding juxta medullary nephron.

- Loop of Henle of juxta medullary nephron runs deep into medulla.
- Juxta medullary nephrons outnumber the cortical nephrons.
- Juxta medullary nephrons are located in the columns of Bertini.
- Renal corpuscle of juxta medullary nephron lies in the outer portion of the renal medulla.

Neural Control and Coordination

(Central Nervous System)

93. Given below are two statements:

Statement I: The cerebral hemispheres are connected by nerve tract known as corpus callosum.

Statement II: The brain stem consists of the medulla oblongata, pons and cerebrum.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

Chemical Coordination and Integration

(Adrenal gland, Pancreas)

94. Match List-I with List-II:

List-I		List-II	
(A)	Exophthalmic goitre	(I)	Excess secretion of cortisol, moon face and hyperglycemia
(B)	Acromegaly	(II)	Hypo-secretion of thyroid hormone and stunted growth
(C)	Cushing's syndrome	(III)	Hyper secretion of thyroid hormone and protruding eye balls.
(D)	Cretinism	(IV)	Excessive secretion of growth hormone

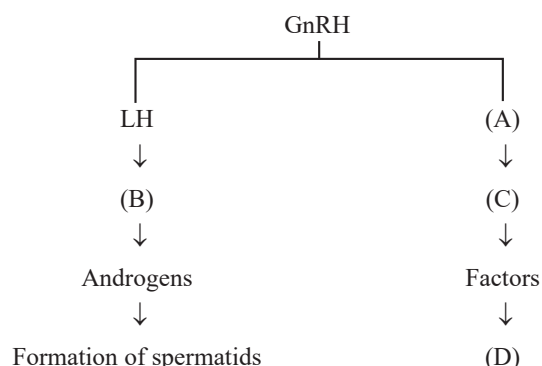
Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(II), D-(I)
- A-(III), B-(IV), C-(I), D-(II)
- A-(I), B-(III), C-(II), D-(IV)
- A-(IV), B-(II), C-(I), D-(III)

Human Reproduction

(Gametogenesis)

95. Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.



- FSH, Sertoli cells, Leydig cells, spermatogenesis.
- ICSH, Leydig cells, Sertoli cells, spermatogenesis.
- FSH, Leydig cells, Sertoli cells, spermiogenesis
- ICSH, Interstitial cells, Leydig cells, spermiogenesis.

Principles of Inheritance and Variation

(Mendal's Laws of Inheritance And Inheritance of One Gene)

96. As per ABO blood grouping system, the blood group of father is B^+ , mother is A^+ and child is O^+ . Their respective genotype can be

- $I^B i / I^A i / ii$
- $I^B I^B / I^A I^A / ii$
- $I^A I^B / i I^A / I^B i$
- $I^A i / I^B i / I^A i$
- $i I^B / i I^A / I^A I^B$

Choose the most appropriate answer from the options given below:

- C & B only
- D & E only
- A only
- B only

Molecular Basis of Inheritance

(Transcription)

97. Match List-I with List-II:

List-I		List-II	
(A)	RNA polymerase III	(I)	snRNPs
(B)	Termination of transcription	(II)	Promotor
(C)	Splicing of Exons	(III)	Rho factor
(D)	TATA box	(IV)	SnRNAs, tRNA

Choose the correct answer from the options given below:

- A-(III), B-(IV), C-(I), D-(II)
- A-(IV), B-(III), C-(I), D-(II)
- A-(II), B-(IV), C-(I), D-(III)
- A-(III), B-(II), C-(IV), D-(I)

Evolution

(Brief Account of Evolution)

98. Match List-I with List-II:

List-I		List-II	
(A)	Mesozoic Era	(I)	Lower invertebrates
(B)	Proterozoic Era	(II)	Fish & Amphibia
(C)	Cenozoic Era	(III)	Birds & Reptiles
(D)	Paleozoic Era	(IV)	Mammals

Choose the correct answer from the options given below:

- A-(I), B-(II), C-(IV), D-(III)
- A-(III), B-(I), C-(IV), D-(II)
- A-(II), B-(I), C-(III), D-(IV)
- A-(III), B-(I), C-(II), D-(IV)

Human Health and Diseases

(Immunity)

99. Given below are two statements:

Statement I: Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.

Statement II: Both bone marrow and thymus provide micro environments for the development and maturation of T-lymphocytes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

Organisms and Populations

(Population Interactions)

100. Given below are two statements :

Statement I: Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.

Statement II: According to Gause's principle during competition, the inferior will be eliminated. This may be true if resources are limiting.

In the light of the above statements, choose correct answer from the options given below:

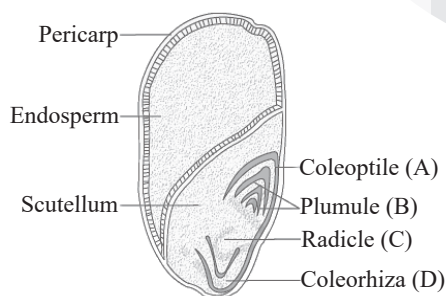
- Statement I is true but Statement II is false.
- Statement I is false but Statement II is true.
- Both Statement I and Statement II are true.
- Both Statement I and Statement II are false.

Answer Key

1. (d)	2. (c)	3. (a)	4. (b)	5. (c)	6. (c)	7. (c)	8. (b)	9. (c)	10. (a)
11. (c)	12. (d)	13. (d)	14. (c)	15. (a)	16. (b)	17. (a)	18. (a)	19. (b)	20. (d)
21. (d)	22. (a)	23. (d)	24. (b)	25. (a)	26. (a)	27. (d)	28. (a)	29. (c)	30. (a)
31. (a)	32. (d)	33. (c)	34. (b)	35. (b)	36. (a)	37. (c)	38. (c)	39. (d)	40. (c)
41. (d)	42. (a)	43. (a)	44. (d)	45. (c)	46. (d)	47. (b)	48. (a)	49. (a)	50. (d)
51. (d)	52. (d)	53. (d)	54. (d)	55. (d)	56. (b)	57. (a)	58. (b)	59. (a)	60. (c)
61. (d)	62. (c)	63. (d)	64. (b)	65. (d)	66. (d)	67. (b)	68. (a)	69. (c)	70. (b)
71. (c)	72. (b)	73. (b)	74. (a)	75. (c)	76. (a)	77. (b)	78. (b)	79. (d)	80. (a)
81. (b)	82. (d)	83. (a)	84. (a)	85. (a)	86. (a)	87. (a)	88. (c)	89. (a)	90. (c)
91. (d)	92. (a)	93. (a)	94. (b)	95. (c)	96. (c)	97. (b)	98. (b)	99. (c)	100. (b)

Explanations

- (d)** The morphology of the mycelium, mode of spore formation and fruiting bodies form the basis for the division of the kingdom fungi into various classes. All fungi derive their nutrition through heterotrophic mode of nutrition.
- (c)** Some common members of basidiomycetes are *Agaricus* (mushroom), *Ustilago* (smut) and *Puccinia* (rust fungus). *Rhizopus* is also known as the bread mould. It belongs to Phycmycetes.
- (a)** The primary root, or radicle is the first organ to appear when a seed germinates.



- (b)** Given figures (A) and (B) shows perigynous flower. In this, gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level. The ovary here is said to be half inferior, e.g., plum, rose, peach.
- (c)** *Datura* is an example of actinomorphic flower as their flowers can be divided into two equal radial halves in any radial plane passing through the centre.
- (c)** In monocots, under water stress condition, bulliform cells become flaccid that makes the leaves curl inwards to minimise water loss.
- (c)** In the given figure of stomata with bean shaped guard cells, A-Epidermal cells, B-Subsidiary cells, C-Guard cells, D-Stomatal pore. The outer walls of guard cells (away from the stomatal pore) are thin and the inner walls (towards the stomatal pore) are highly thickened.
- (b)** Parenchyma and collenchyma are living tissue. Gymnosperms lack vessels in their xylem but in flowering plants, tracheids and vessels are the main water transporting elements.
- (c)**
 - Nucleolus - site for active ribosomal RNA synthesis.
 - Centriole - organisation like cartwheel
 - Leucoplasts - for storing nutrients
 - Golgi apparatus- site for formation of glycolipid
- (a)** Inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition. Malonate resembles the substrate succinate and competes with the substrate for the substrate-binding site of the enzyme. Consequently, the substrate cannot bind and as a result, the enzyme action declines.
- (c)** Metal ions are the common cofactors. Cofactor for enzyme carboxypeptidase is zinc.
- (d)** Lecithin or phosphatidyl choline, a small molecular weight organic compound found in living tissues, is an example of phospholipid.
- (d)** Spindle fibres attach to kinetochores of chromosomes during metaphase stage. Chromosomes are moved to spindle equator and get aligned along metaphase plate through spindle fibres to both poles.

14. (c) During leptotene stage the chromosomes become gradually visible under the light microscope.

The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers.

15. (a) For the dark reaction or Calvin cycle, CO_2 is required for the carboxylation step and ATP and NADH are required for the reduction and regeneration step.

16. (b) For every CO_2 molecule entering the Calvin cycle, 3 molecules of ATP and 2 of NADPH are required.

The Calvin cycle can be described under three stages: carboxylation, reduction and regeneration.

Reduction step involve utilisation of 2 molecules of ATP for phosphorylation and two of NADPH for reduction per CO_2 molecule fixed. The regeneration steps require one ATP for phosphorylation to form RuBP.

17. (a) Plants show an interesting phenomenon, the living differentiated cells, that by now have lost the capacity to divide can regain the capacity of division under certain conditions. This phenomenon is termed as dedifferentiation. For example, formation of meristems – interfascicular cambium and cork cambium from fully differentiated parenchyma cells.

18. (a) Auxins are widely used as herbicides. 2, 4-D, widely used to kill dicotyledonous weeds, does not affect mature monocotyledonous plants.

19. (b) *Vallisneria* is a water pollinated flowers which are not very colourful and do not produce nectar.

In most of the water-pollinated species, pollen grains are protected from wetting by a mucilaginous covering.

In water lily, the flowers emerge above the level of water and are pollinated by insects or wind.

Pollen grains in many water pollinated plants like seagrasses are long, ribbon like and they are carried passively inside the water; some of them reach the stigma and achieve pollination.

20. (d) To determine the genotype of the black seed plant, a test cross is performed with a homozygous recessive (bb) individual. This cross will reveal whether the black seed plant is homozygous dominant (BB) or heterozygous (Bb) based on the offspring's phenotypes.

21. (d)

Pink flowered snapdragon plant (Rr)		×	Red flowered snapdragon plant (RR)	
Pink flower →			R	r
Red flower →	R		RR	Rr
	r		Rr	rr

50% flowers are red in colour and 50% flowers are pink in colour.

22. (a) Two or more alternative forms of a gene- Allele

Cross of F_1 progeny with homozygous recessive parent- Test cross

Cross of F_1 progeny with any of the parents- Back cross

Number of chromosome sets in plant- Ploidy

23. (d) Alleles show expressions in the F_2 generation.

24. (b) A transcription unit in DNA is defined primarily by the three regions in the DNA:

(i) A Promoter

(ii) The Structural gene

(iii) A Terminator

25. (a) In the absence of a preferred carbon source such as glucose, if lactose is provided in the growth medium of the bacteria, the lactose is transported into the cells through the action of permease.

26. (a) • *Clostridium butylicum* - Butyric acid

• *Saccharomyces cerevisiae*- Ethanol

• *Trichoderma polysporum*- Cyclosporin-A

• *Streptococcus* sp.- Streptokinase

27. (d) Recognition sequence of first restriction endonuclease *Hind* II is of 6 base pairs and it generates blunt ends.

28. (a) When a piece of DNA carrying gene of interest is transferred into an alien organism – it may get integrated itself into the genome of the recipient and may multiply and be inherited along with the host DNA.

29. (c) The capacity to generate a whole plant from any cell of the plant is called totipotency.

30. (a) *Bt* toxins are insect-group specific. The toxin is coded by a gene *cryI*Ac named *cry*.

The *Bt* toxin protein exist as inactive protoxins but once an insect ingest the inactive toxin, it is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals.

31. (a) Verhulst-Pearl Logistic Growth is described by the following equation:

$$dN / dt = rN \left(\frac{K - N}{K} \right)$$

Where N = Population density at time t

r = Intrinsic rate of natural increase

K = Carrying capacity

32. (d) *Ex situ* Conservation– In this approach, threatened animals and plants are taken out from their natural habitat and placed in special setting where they can be protected and given special care. Zoological parks, botanical gardens and wildlife safari parks serve this purpose.

33. (c) Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification.

Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity.

There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity.

34. (b) Causes of biodiversity losses: The accelerated rates of species extinctions that the world is facing now are largely due to human activities. There are four major causes (‘The Evil Quartet’ is the sobriquet used to describe them).

1. Habitat loss and fragmentation
2. Over-exploitation
3. Alien species invasions
4. Co-extinctions

35. (b) The IUCN Red List (2004) documents the extinction of 784 species (including 338 vertebrates, 359 invertebrates and 87 plants) in the last 500 years.

36. (a) Sexual reproduction can be isogamous, anisogamous or oogamous in phaeophyceae.

37. (c)

Rose	Perigynous flower
Pea	Marginal placentation
Cotton	Twisted aestivation
Mango	Drupe

38. (c)

Monoadelphous	China rose
Diadelphous	Pea
Polyadelphous	Citrus
Epiphyllous	Lily

39. (d) Chloroplast contains small, double-stranded circular DNA molecules and ribosomes.

40. (c) • GLUT-4 enables glucose transport into cells.

- Insulin is a hormone secreted by pancreas.
- Trypsin is an example of enzyme.
- Collagen is the most abundant protein in animal world, which is an intercellular ground substances.

41. (d) Sugarcane stores carbohydrate as sugar in their stems. Spraying sugarcane crop with gibberellins increases the length of the stem, thus increasing the yield by as much as 20 tonnes per acre.

42. (a) In C_3 plants some O_2 does bind to RuBisCO, and hence CO_2 fixation is decreased. In C_4 plants photorespiration does not occur.

43. (a) The step in the tricarboxylic acid cycle that does not involve oxidation of substrate is

Succinyl-CoA $\rightarrow$ Succinic acid

This step involves the conversion of Succinyl-CoA to Succinic Acid by cleavage of CoA from Succinyl-CoA, which is a substrate-level phosphorylation, not an oxidation step.

44. (d)

Citric acid cycle	Mitochondrial matrix
Glycolysis	Cytoplasm
Electron transport system	Inner mitochondrial membrane
Proton gradient	Intermembrane space of mitochondria

45. (c) The given diagram shows a wind-pollinated plant showing compact inflorescence and well exposed stamens.



46. (d)

Frederick Griffith	Transformation
Francois Jacob and Jacques Monod	Lac operon
Har Gobind Khorana	Genetic code
Meselson and Stahl	Semi-conservative mode of DNA replication

47. (b) The DNA-dependent DNA polymerase in *E. coli* catalyse polymerisation only in one direction, that is $5' \rightarrow 3'$.

48. (a) Protoplasts fused in somatic hybridisation involving two varieties of plants.

49. (a) The NPP of the first trophic level is the GPP of the second trophic level and the NPP of the 2nd trophic level is GPP is 3rd trophic level.

According to the 10% law 90 per cent is the respiratory loss (R) at each trophic level.

$$NPP = GPP - R$$

At the first trophic level NPP is 100x which is GPP of second trophic level.

At second trophic level NPP is 10% of GPP = 10x

Since the GPP of the 3rd trophic level is the same as the NPP of the second trophic level, it is equal to 10x.

50. (d)

Robert May	Global species diversity at about 7 million
Alexander von Humboldt	Species-Area relationship
Paul Ehrlich	Rivet popper hypothesis
David Tilman	Long term ecosystem experiment using outdoor plots

51. (d)

Annelids – True coelomates

Poriferans to Platyhelminthes – Acoelomates

Aschelminthes – Pseudocoelomates

52. (d) *Pleurobrachia* is the example of phylum ctenophora.

Radula is a file like rasping organ found in phylum mollusca.
Stomochord is found in phylum hemichordata similar to notochord of chordata.

Air - bladder maintains buoyancy in Osteichthyes.

53. (d) Scientific name of Angel fish is *Pterophyllum*.

Myxine is also known as hagfish.

Scientific name of saw fish is *Pristis*.

Scientific name of flying fish is *Exocoetus*.

54. (d)

Skeletal muscles: Triceps

Smooth muscles: Stomach

Cardiac muscles: Heart

55. (d) A pair of jointed filamentous structures called anal cerci are present on the 10th abdominal segment in both sexes of cockroaches.

56. (b) The core of cilia/flagella is called axoneme.

Centrioles have a '9+0' structure of triplet microtubules arranged in a cartwheel pattern.

Cristae are folds in the inner mitochondrial membrane.

Satellite is the term given to the end of a chromosome that is separated from the rest of the chromosome by a secondary constriction.

57. (a)

	List-I		List-II
A.	Lipase	II.	Ester bond
B.	Nuclease	IV.	Phosphodiester bond
C.	Protease	I.	Peptide bond
D.	Amylase	III.	Glycosidic bond

58. (b) The correct sequence of stages of cell division is:
Gap-1 phase → Synthesis phase → Gap-2 phase → Karyokinesis → Cytokinesis

59. (a)

Diakinesis	-	Completion of terminalisation of chiasmata
Pachytene	-	Appearance of recombination nodules
Zygotene	-	Synaptonemal complex formation
Leptotene	-	Chromosomes look like thin threads

60. (c)

	List-I		List-II
A.	Expiratory capacity	II.	Tidal Volume + Expiratory Reserve Volume
B.	Functional residual capacity	IV.	Expiratory Reserve Volume + Residual Volume
C.	Vital capacity	I.	Tidal Volume + Expiratory Reserve Volume + Inspiratory Reserve Volume
D.	Inspiratory capacity	III.	Tidal Volume + Inspiratory Reserve Volume

61. (d) Factors which are favourable for formation of oxyhaemoglobin in the alveoli – High pO_2 , high pH, low pCO_2 , lower temperature, lesser H^+ concentration.

62. (c) Correct sequence is –

SA node (E) → AV node (C) → AV bundle (A) →

Bundle branches (D) → Purkinje fibres (B).

63. (d) Descending limb of Henle's loop - Permeable to water and impermeable to electrolytes.

Proximal convoluted tubule - Lined by simple cuboidal brush border epithelium.

64. (b)

List-I	List-II
Fibrous joints	Seen in flat skull bones and do not allow any movement.
Cartilaginous joints	Present between adjacent vertebrae and allow limited movements.
Hinge joints	Present in knees and help in locomotion.
Ball and socket joints	Present between humerus and pectoral girdle and allow rotational movement.

65. (d) Myasthenia gravis, Rheumatoid arthritis and Systemic Lupus Erythematosus are autoimmune disorders.

Gout is not an autoimmune disorder; it results from uric acid crystal accumulation. Muscular dystrophy is a genetic disorder, not autoimmune.

66. (d)

	List-I		List-II
A.	Pons	III.	Connects different regions of the brain
B.	Hypothalamus	IV.	Neurosecretory cells
C.	Medulla	II.	Controls respiration and gastric secretions
D.	Cerebellum	I.	Provides additional space for neurons, regulates posture and balance

67. (b) Glucagon is a peptide hormone secreted by the alpha cells of Islets of Langerhans of pancreas.

Remaining are steroidal hormones.

68. (a) The opening of the vagina is often covered partially by a membrane called hymen. The hymen is often torn during the first coitus (intercourse).

However, it can also be broken by a sudden fall or jolt, insertion of a vaginal tampon, active participation in some sports like horseback riding, cycling, etc. In some women the hymen persists even after coitus. Therefore, the presence or absence of hymen is not a reliable indicator of virginity or sexual experience.

69. (c) Uterine fundus is a part of uterus. Infundibulum, ampulla and isthmus are the parts of fallopian tube.

70. (b) FSH acts upon ovarian follicles in females and Sertoli cells in males.

71. (c) Breast feeding during initial periods of infants growth is recommended by doctors for bringing a healthy baby because colostrum contains several antibodies absolutely essential for develop resistance for new born.

72. (b)

	List-I		List-II
A.	Non-medicated IUD	III	Lippes loop
B.	Copper releasing IUD	I	Multiload 375
C.	Hormone releasing IUD	IV	LNG-20
D.	Implants	II	Progestogens

73. (b) Vaults are barrier methods of contraception while Lactational amenorrhea, Coitus interruptus, Periodic abstinence are natural/traditional method of contraception.

74. (a) Down's syndrome - 21st chromosomes
 α -thalassemia - 16th chromosomes
 β -thalassemia - 11th chromosomes
 Klinefelter's syndrome - 'X' chromosome

75. (c) The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

The process of transcription is catalyzed by the enzyme DNA dependent RNA polymerase in only one direction, that is, 5' → 3', the strand of DNA that has the polarity 3' → 5' acts as a template, and is also referred to as template strand.

The given template strand of DNA is:

3'TACATGGCAAATATCCATTCA5'

So, the transcribed strand of RNA will be:

5'AUGUACCGUUUAUAGGUAAGU3'.

76. (a) The flippers of Penguins and Dolphins are analogous organs and thus are examples of convergent evolution, where different species have independently evolved similar traits to adapt to similar environments, like swimming in water.

77. (b) Gene pool (total number of genes and alleles in a population) will remain constant if a population is following Hardy-Weinberg equilibrium. The factors affecting Hardy-Weinberg equilibrium are: genetic drift, mutation, gene migration or gene flow, genetic recombination and natural selection.

78. (b) *Homo habilis* (A) → *Homo erectus* (D) →
Homo neanderthalensis (C) → *Homo sapiens* (B).

79. (d)

	List-I		List-II
A.	Typhoid	IV.	Bacteria
B.	Leishmaniasis	III.	Protozoa
C.	Ringworm	I.	Fungus
D.	Filariasis	II.	Nematode

80. (a) Common cold is caused by rhinoviruses.

Haemozoin is toxic substance released from ruptured RBCs which were infected by malarial protozoa (*Plasmodium*).

Widal test is used for diagnosis of typhoid.

Allergy is exaggerated response of immune system caused by allergens (Dust mites etc).

81. (b)

List-I	List-II
Cocaine	Obtained from coca plant (<i>Erythroxylum coca</i>)
Heroin	<i>Papaver somniferum</i>

Morphine	Effective and sedative painkiller; very useful in patients who have undergone surgery.
Marijuana	Flower tops, leaves and the resin of <i>Cannabis sativa</i>

82. (d) X – *ori* and Y – *rop*

ori responsible for controlling copy number of the linked DNA.

rop codes for the proteins involved in the replication of the plasmid.

83. (a) The 'Ti plasmid' of *Agrobacterium tumefaciens* stands for Tumor inducing plasmid.

84. (a) Bio-reactors are used to produce large scale bacterial culture.

85. (a)

List-I	List-II
α -1-antitrypsin	protein is used to treat emphysema.
<i>cryIAb</i>	controls corn borer
<i>cryIAC</i>	cotton bollworms.
Enzyme replacement therapy	treat ADA deficiency.

86. (a) Notochord is absent in non-chordate instead it is a characteristic of chordate. In non-chordate heart is dorsal (if present) but in chordate it is ventral. Post anal tail is found in chordate. Pharynx is perforated by gill slits in chordates. Central nervous system is dorsal in chordates whereas ventral in non-chordates.

87. (a) Goblet cells of alimentary canal is an example of unicellular glandular epithelium. Compound epithelium is found lining both dry surfaces, such as skin, and moist surfaces, such as the buccal cavity.

Salivary gland is an example of multicellular glandular epithelium.

Pancreas is a type of endocrine glandular epithelium. It secretes insulin and glucagon hormone.

88. (c) The structure used for storing food in cockroaches is Crop.

A ring of 6-8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and midgut, which secrete digestive juice.

At the junction of midgut and hindgut is present another ring of 100-150 yellow coloured thin filamentous Malpighian tubules.

Gizzard helps in grinding the food particles in cockroach.

89. (a) Like mitochondria, the chloroplasts are also double membrane bound. Of the two, the inner chloroplast membrane is relatively less permeable.

90. (c) The correct sequential steps in the enzyme action is as follows:

E. Substrate binding to active site.

A. Substrate enzyme complex formation.

D. Chemical bonds of the substrate broken.

C. Release of products.

B. Free enzyme ready to bind with another substrate.

91. (d) In ECG, The P-wave represents the electrical excitation (or depolarisation) of the atria, which leads to the contraction of both the atria.

The QRS complex represents the depolarisation of the ventricles, which initiates the ventricular contraction.

The T-wave represents the return of the ventricles from excited to normal state (repolarisation). The end of the T-wave marks the end of systole.

T-P gap represents joint diastole of heart, in which heart muscles are electrically silent.

92. (a) In juxtamedullary nephron, the loop of Henle is very long and runs deep into the medulla.

Cortical nephron is present in majority w.r.t. juxtamedullary nephron.

93. (a) Three major regions make up the brain stem; mid brain, pons and medulla oblongata. Cerebrum is the part of forebrain.

94. (b) Exophthalmic goitre is a form of hyperthyroidism, characterised by enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate, and weight loss, also called Graves' disease.

Acromegaly is the excessive secretion of growth hormone.

Cushing's syndrome is caused by excessive secretion of cortisol.

Cretinism is the hyposecretion of thyroid hormone during pregnancy which leads to stunted growth (cretinism), mental retardation, low intelligence quotient, abnormal skin, deaf-mutism, etc.

95. (c)

A - FSH

B - Leydig cell

C - Sertoli cells

D - Spermiogenesis

The increased levels of GnRH acts at the anterior pituitary gland and stimulates secretion of two gonadotropins – luteinising hormone (LH) and follicle stimulating hormone (FSH). LH acts at the Leydig cells and stimulates synthesis and secretion of androgens. Androgens, in turn, stimulate the process of spermatogenesis. FSH acts on the Sertoli cells and stimulates secretion of some factors which help in the process of spermiogenesis.

96. (c) If a child has blood group O^+ (ii) so according to that both the parents should have at least one recessive allele (i). So the correct answer is father ($I^B i$) and the mother is ($I^A i$).

97. (b)

A. RNA Polymerase III - SnRNAs, tRNA

B. Termination of transcription - Rho factor

C. Splicing of exons - snRNPs

D. TATA box - Promoter

98. (b) Mesozoic Era - Birds and reptiles

Proterozoic Era - Lower invertebrates

Cenozoic Era - Mammals

Paleozoic Era - Fish and amphibians

99. (c) The primary lymphoid organs include both the bone marrow and the thymus. All blood cells, including lymphocytes, are produced in the bone marrow. Both the bone marrow and thymus offer micro environments crucial for the development and maturation of T-lymphocytes.

100. (b) Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.